
Wireless Communications at the Physical Layer

Part 1: Fundamentals

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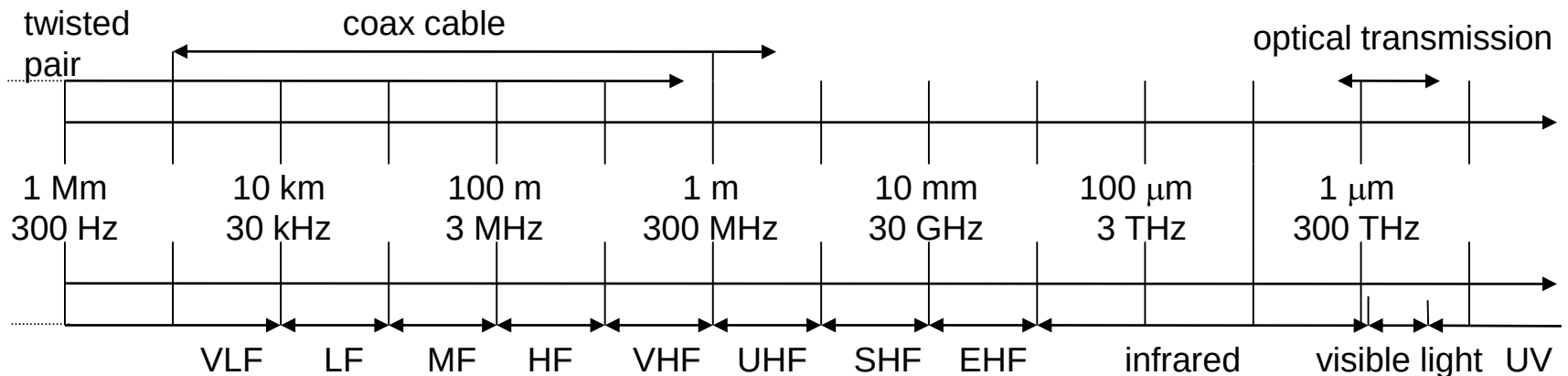
Outline

- Electromagnetic spectrum and allocation
- Frequency domain representation of signals
- Decibel notation
- Antennas
- Signal propagation modes
- Wireless transmission deterioration factors
- Signal-to-noise ratios

Electromagnetic spectrum and allocation

Frequencies for communication

- VLF = Very Low Frequency
 - LF = Low Frequency
 - MF = Medium Frequency
 - HF = High Frequency
 - VHF = Very High Frequency
 - UHF = Ultra High Frequency
 - SHF = Super High Frequency
 - EHF = Extremely High Frequency
 - UV = Ultraviolet Light
- Frequency and wavelength
 - $\lambda = c/f$
 - wavelength λ , speed of light $c \cong 3 \times 10^8 \text{ m/s}$, frequency f



Frequency allocation

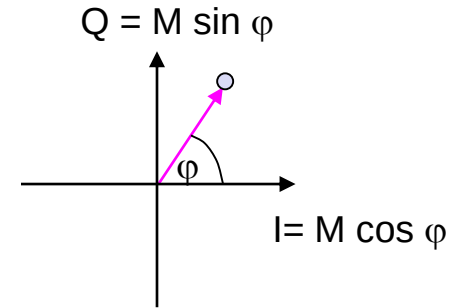
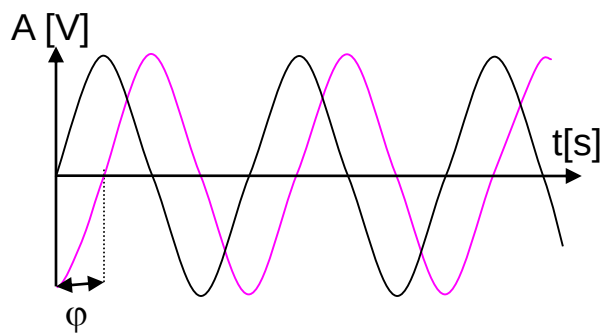
- Some frequencies are allocated to specific uses
 - Cellular phones, analog television/radio broadcasting, DVB-T, radar, emergency services, radio astronomy, ...
- Particularly interesting: **ISM bands** ("Industrial, scientific, medical") – license-free operation
- Quadro Nacional de Atribuição de Frequências
 - <http://www.anacom.pt/render.jsp?categoryId=290215>

Some typical ISM bands	
Frequency	Comment
13,553-13,567 MHz	
26,957 – 27,283 MHz	
40,66 – 40,70 MHz	
433 – 464 MHz	Europe
900 – 928 MHz	Americas
2,4 – 2,5 GHz	WLAN/WPAN
5,725 – 5,875 GHz	WLAN
24 – 24,25 GHz	

Frequency domain representation of signals

Signals

- Different representations of signals
 - time (time domain)
 - frequency spectrum (frequency domain)
 - phase state diagram (amplitude M and phase φ in polar coordinates)



- Composed signals translated into frequency domain using Fourier transform
- Digital signals need
 - Modulation with a carrier frequency for transmission

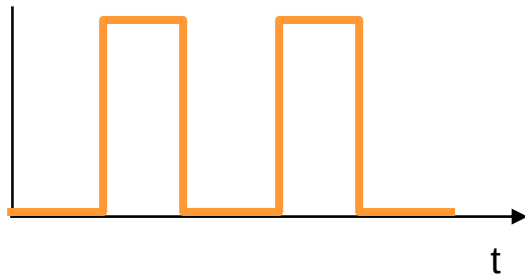
Fourier representation of periodic signals

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

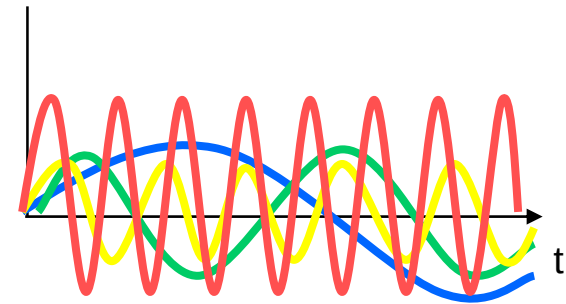
$c/2$ - DC component of the signal

$n = 1$ - fundamental

$n = 2, 3, \dots$ - harmonics

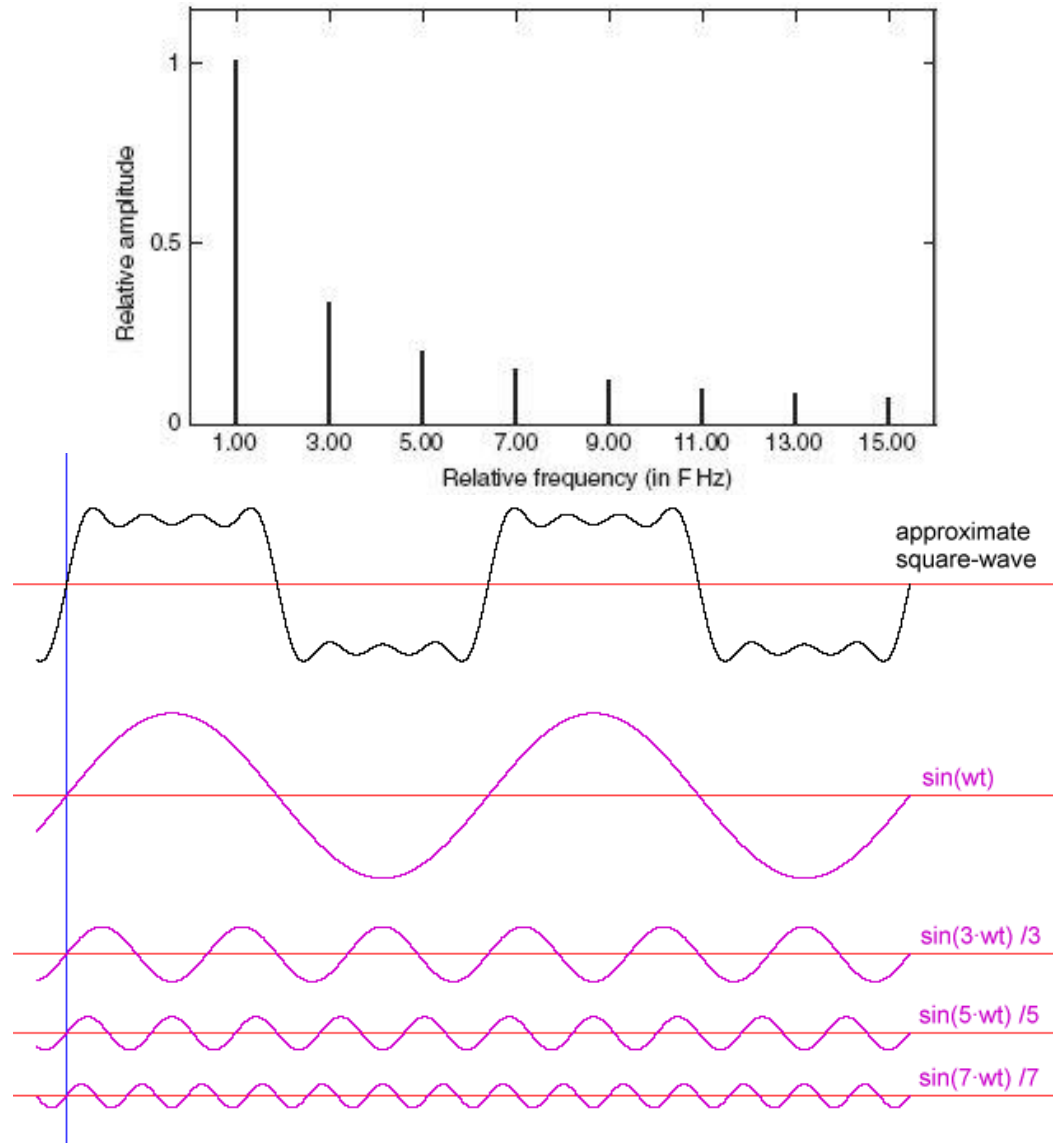


ideal periodic signal



real composition
(based on harmonics)

Example: square wave



Decibel notation

Decibel Notation (dB)

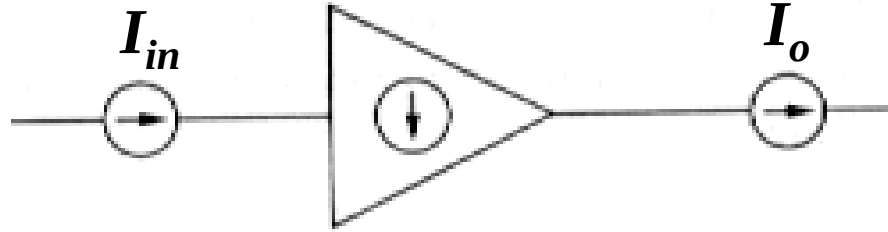
- Logarithmic ratio of two signal levels

$$\text{dB (Decibel)} = 10 \log_{10} (P_2/P_1)$$

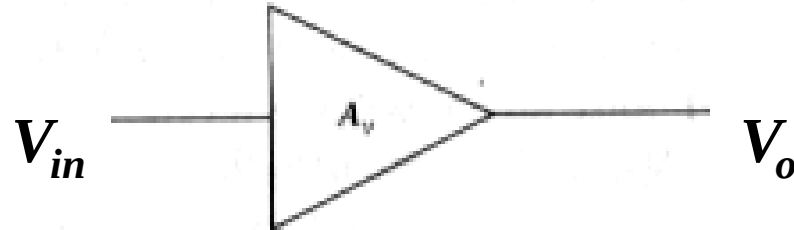
- A gain of **3 dB** represents **twice** the power
- A **gain** of **-3 dB** (**loss** of 3 dB) represents **half** of the power
- A gain of **10 dB** represents **10 times** the power
- A gain of **20 dB** represents **100 times** the power

Basic equations to calculate decibels

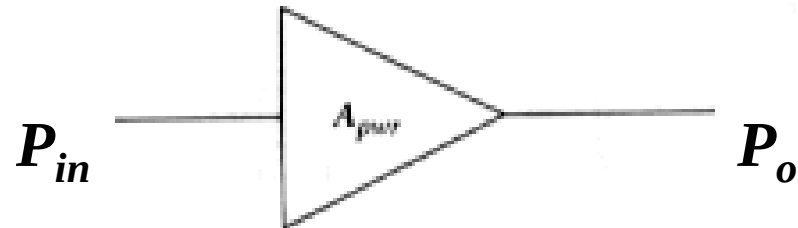
$$dB = 20 \log \left(\frac{I_o}{I_{in}} \right)$$



$$dB = 20 \log \left(\frac{V_o}{V_{in}} \right)$$



$$dB = 10 \log \left(\frac{P_o}{P_{in}} \right)$$



Special decibel scales

- **dBm (dB miliWatt)**
- Power Relative to 1 mW, i.e. 0 dBm is 1 mW

$$P[\text{dBm}] = 10 \log (P[\text{W}]/1 \text{ mW})$$

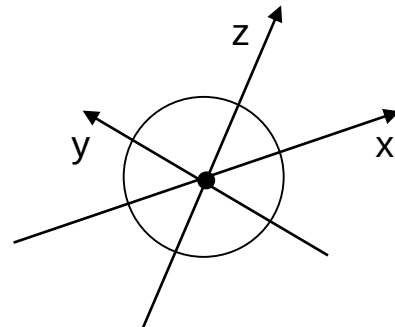
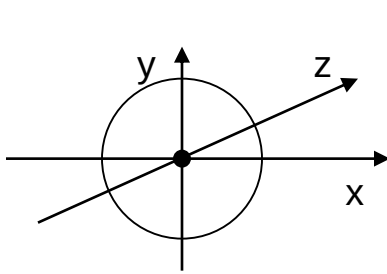
- Very used in radiofrequency measurements (RF)
- Ex: What is the signal level 9 mW as expressed in dBm?
 - $P[\text{dBm}] = 10 \log (9 \text{ mW}/1 \text{ mW}) = 9.54 \text{ dBm}$

- **dBW (dB Watt)**
- Power Relative to 1 W, i.e. 0 dBW is 1 W

Antennas

Antennas

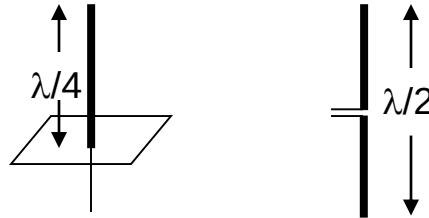
- For radiation and reception of electromagnetic waves
- In two-way communication, the same antenna can be used for transmission and reception
- Isotropic radiator: equal radiation in all directions (three dimensional) - only a theoretical reference antenna
- Real antennas always have directive effects (vertically and/or horizontally)
- Radiation pattern: measurement of radiation around an antenna



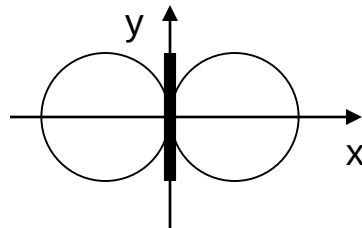
ideal
isotropic
radiator

Antennas: dipoles

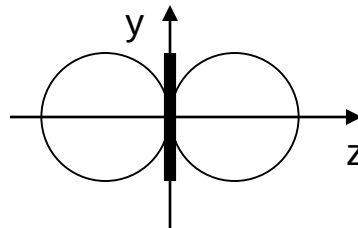
- Real antennas are not isotropic radiators but, e.g., dipoles with lengths $\lambda/4$ on car roofs or $\lambda/2$ as Hertzian dipole
→ size of antenna proportional to wavelength



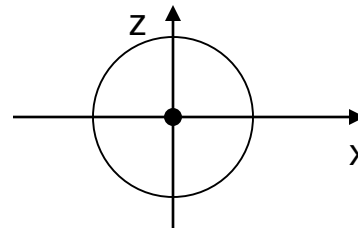
- Example: Radiation pattern of a simple Hertzian dipole



side view (xy-plane)



side view (yz-plane)



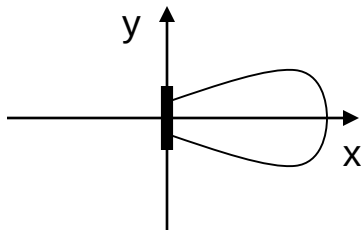
top view (xz-plane)

simple
dipole

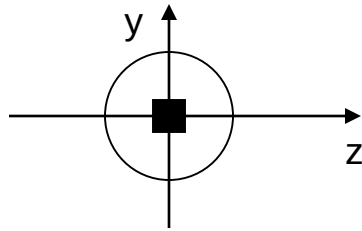
- Gain: maximum power in the direction of the main lobe compared to the power of an isotropic radiator (with the same average power)

Antennas: directed and sectorized

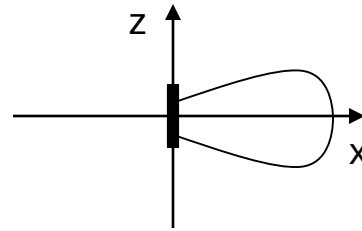
- Often used for microwave connections or base stations for mobile phones (e.g., radio coverage of a valley)



side view (xy-plane)

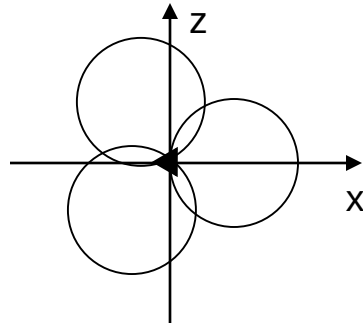


side view (yz-plane)

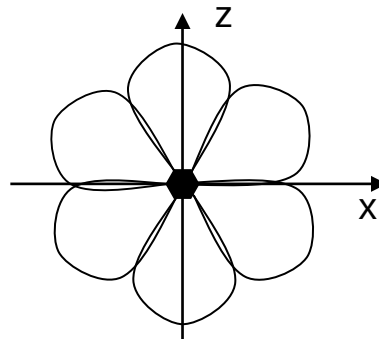


top view (xz-plane)

directed
antenna



top view, 3 sector



top view, 6 sector

sectorized
antenna

Antenna Gain

- Antenna gain
 - A measure of the directionality of an antenna.
 - Relation of power output, in a particular direction, compared to that produced by omnidirectional antenna (isotropic)
- Effective area
 - Equivalent area from which an antenna directed toward the source of a electromagnetic wave absorbs its energy
- Relationship between antenna gain and effective area

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

- G = antenna gain
- A_e = effective area
- f = carrier frequency
- c = speed of light ($\sim 3 \times 10^8$ m/s)
- λ = carrier wavelength

Antenna gain and effective area

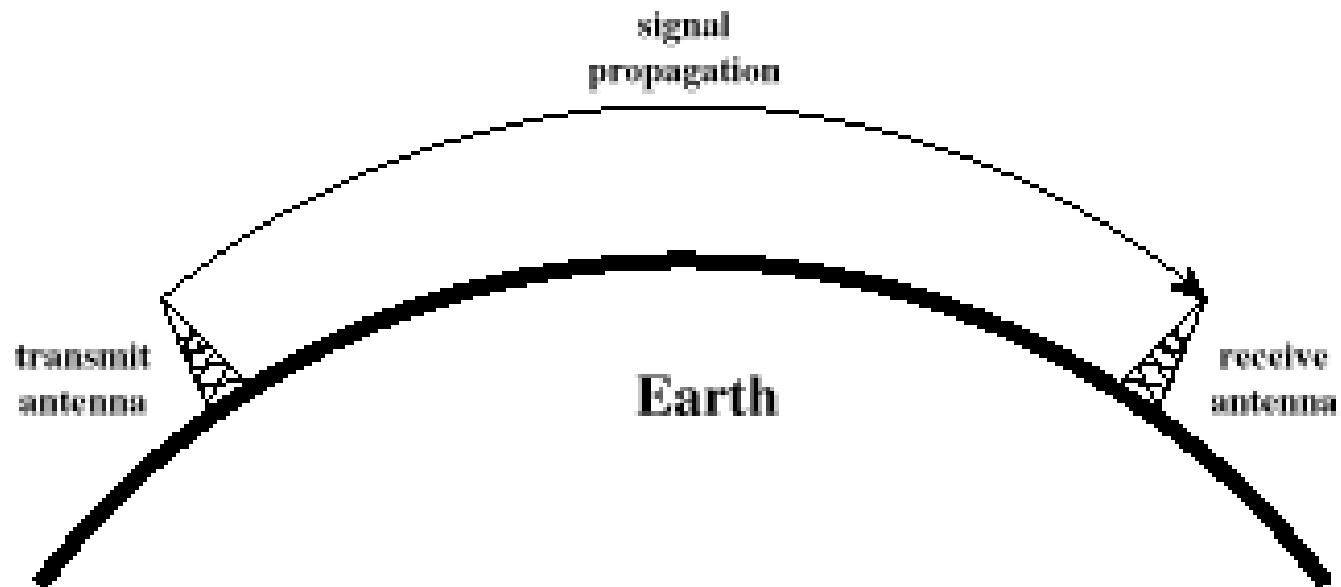
Type of antenna	Effective area	Power gain
Isotropic	$\lambda^2/4\pi$	1
Infinitesimal dipole or loop	$1.5\lambda^2/4\pi$	1.5
Half-wave dipole	$1.64\lambda^2/4\pi$	1.64
Horn, mouth area A	$0.81A$	$10A/\lambda^2$
Parabolic, face area A	$0.56A$	$7A/\lambda^2$
turnstile	$1.15\lambda^2/4\pi$	1.15

-
- Ground-wave propagation
 - Sky-wave propagation
 - Line-of-sight propagation

Signal propagation modes

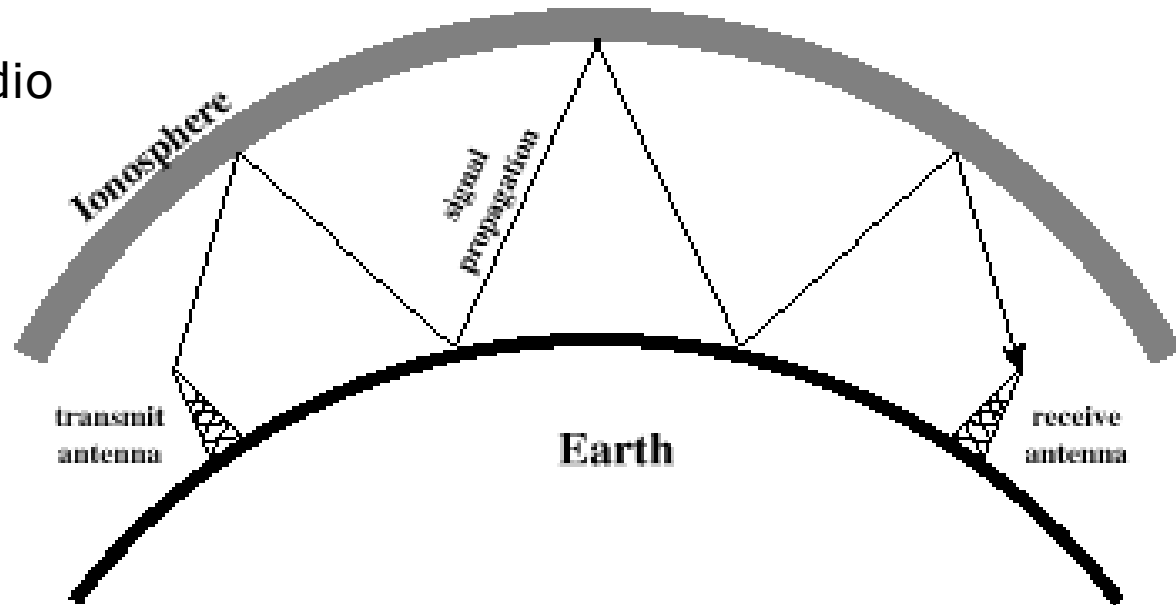
Ground Wave Propagation

- Follows contour of the earth
- Can Propagate considerable distances
- Frequencies up to 2 MHz
- Example
 - AM radio



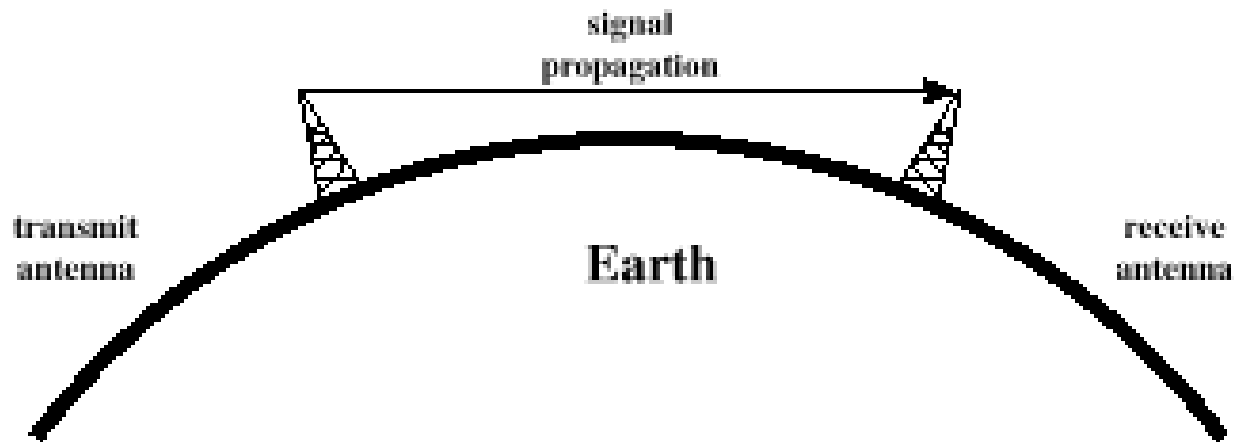
Sky Wave Propagation

- Signal reflected from ionized layer of atmosphere back down to earth
- Signal can travel a number of hops, back and forth between ionosphere and earth's surface
- Frequencies around 2 to 30 MHz
- Reflection effect caused by refraction
- Examples
 - Amateur radio
 - CB radio



Line-of-Sight (LOS) Propagation

- Transmitting and receiving antennas must be within line of sight
 - Satellite communication – signal above 30 MHz not reflected by ionosphere
 - Ground communication – antennas within *effective* line of sight of each other due to refraction
- Refraction – bending of microwaves by the atmosphere
 - Velocity of electromagnetic wave is a function of the density of the medium
 - When wave changes medium, speed changes
 - Wave bends at the boundary between mediums



Line-of-Sight Equations

- Optical line of sight

$$d = 3.57\sqrt{h}$$

- Effective, or radio, line of sight

$$d = 3.57\sqrt{Kh}$$

- d = distance between antenna and horizon (km)
- h = antenna height (m)
- K = adjustment factor to account for refraction, rule of thumb $K = 4/3$

Line-of-Sight Equations

- Maximum distance between two antennas for LOS propagation:

$$3.57 \left(\sqrt{K h_1} + \sqrt{K h_2} \right)$$

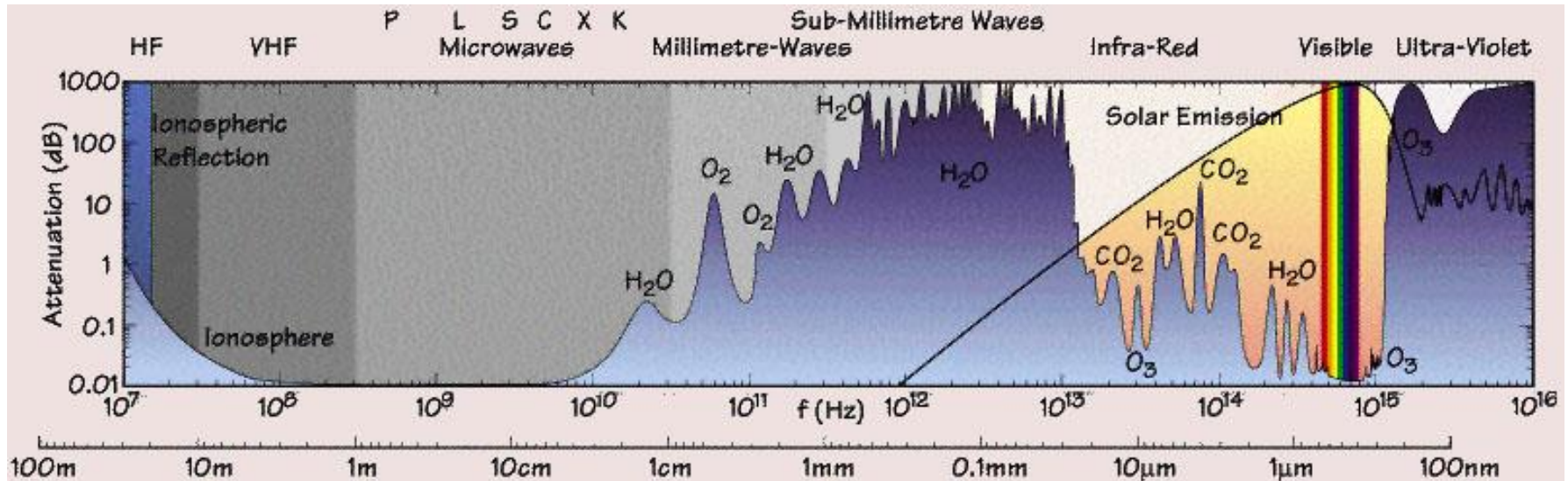
- h_1 = height of antenna one
- h_2 = height of antenna two

-
- Attenuation
 - Free space loss
 - Noise
 - Interference
 - Multipath

Wireless transmission deterioration factors

Attenuation

- Strength of signal (electromagnetic radiation) falls off with distance over transmission medium **due to absorption or scattering of photons**
- Absorption may vary according to medium/material and frequency



- Attenuation/path loss factors for unguided media
 - Received signal must have sufficient strength so that circuitry in the receiver can interpret the signal
 - Signal must maintain a level sufficiently higher than noise to be received without error

Free space loss

- Free space loss, ideal isotropic antenna

$$\frac{P_t}{P_r} = \frac{(4\pi d)^2}{\lambda^2} = \frac{(4\pi f d)^2}{c^2}$$

- P_t = signal power at transmitting antenna
 - P_r = signal power at receiving antenna
 - λ = carrier wavelength
 - d = propagation distance between antennas
 - c = speed of light
- where d and λ are in the same units (e.g., meters)

Free space loss

- Free space loss equation can be recast:

$$\begin{aligned} L_{dB} &= 10 \log \frac{P_t}{P_r} = 20 \log \left(\frac{4\pi d}{\lambda} \right) \\ &= -20 \log(\lambda) + 20 \log(d) + 21.98 \text{ dB} \\ &= 20 \log \left(\frac{4\pi f d}{c} \right) = 20 \log(f) + 20 \log(d) - 147.56 \text{ dB} \end{aligned}$$

* L_{dB} is the **path loss**, in dB

Free space loss

- Free space loss accounting for gain of antennas

$$\frac{P_t}{P_r} = \frac{(4\pi)^2 (d)^2}{G_r G_t \lambda^2} = \frac{(\lambda d)^2}{A_r A_t} = \frac{(cd)^2}{f^2 A_r A_t}$$

- G_t = gain of transmitting antenna
- G_r = gain of receiving antenna
- A_t = effective area of transmitting antenna
- A_r = effective area of receiving antenna

Free space loss equation

- Describes signal strength at distance d relative to some reference distance $d_0 < d$ for which strength is known
- $d_0 \geq \text{far-field distance}$ (depends on antenna technology)
- $L \geq 1$ accounts for system loss

$$\begin{aligned} P_{\text{recv}}(d) &= \frac{P_{\text{tx}} \cdot G_t \cdot G_r \cdot \lambda^2}{(4\pi)^2 \cdot d^2 \cdot L} \\ &= \frac{P_{\text{tx}} \cdot G_t \cdot G_r \cdot \lambda^2}{(4\pi)^2 \cdot d_0^2 \cdot L} \cdot \left(\frac{d_0}{d}\right)^2 = P_{\text{recv}}(d_0) \cdot \left(\frac{d_0}{d}\right)^2 \end{aligned}$$

* $P_{\text{recv}} \Leftrightarrow P_r$

$P_{\text{recv}}(d)$ – Power received (in W) at distance d

$P_{\text{recv}}(d_0)$ – Power received (in W) at distance d_0

Path loss equation

- Generalization of the free space loss equation, to take into account stronger loss than only caused by distance (e.g., walls absorption/reflexion), normally use a larger exponent ($\gamma > 2$)
 - γ is the **path-loss exponent**

$$P_{\text{recv}}(d) = P_{\text{recv}}(d_0) \cdot \left(\frac{d_0}{d}\right)^\gamma$$

- Knowing that:

$$P_{\text{recv}}(d)[\text{dBW}] = P_t[\text{dBW}] - PL(d)[\text{dB}]$$

- Rewrite in logarithmic form (in dB):

$$PL(d)[\text{dB}] = PL(d_0)[\text{dB}] + 10\gamma \log_{10} \left(\frac{d}{d_0}\right)$$

* $PL(d)[\text{dB}] \Leftrightarrow L_{dB}$ is the **path loss**, in dB, at distance d

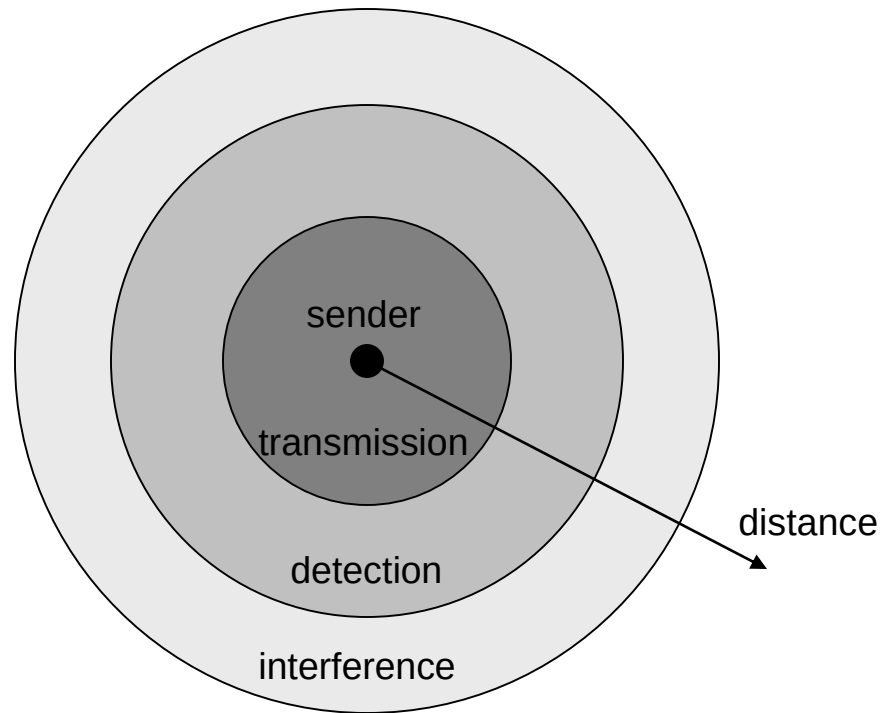
Path loss equation (cont.)

- Take shadowing/fading into account by a random variation
 - Add a Gaussian random variable to dB representation
 - with 0 mean, variance σ^2
 - Equivalent to multiplying with a lognormal distributed random variable in metric units => *lognormal fading*

$$PL(d)[\text{dB}] = PL(d_0)[\text{dB}] + 10\gamma \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma[\text{dB}]$$

Signal propagation ranges

- Transmission range
 - communication possible
 - low error rate
- Detection range
 - detection of the signal possible
 - no communication possible
- Interference range
 - signal may not be detected
 - signal adds to the background noise



Thermal noise

- Type of noise caused due to agitation of electrons
- Present in all electronic devices and transmission media
- Function of temperature
- Cannot be eliminated
- Amount of thermal noise to be found in a bandwidth of 1Hz in any device or conductor is:

$$N_0 = kT \text{ (W/Hz)}$$

- N_0 = noise power density in watts per 1 Hz of bandwidth
- k = Boltzmann's constant = 1.3803×10^{-23} J/K
- T = temperature, in kelvins * (absolute temperature)

* $[K] = [^{\circ}C] + 273.15$

Thermal noise

- Thermal noise is assumed to be independent of frequency (white)
- Thermal noise present in a bandwidth of B Hertz (in watts):

$$N = kTB$$

or, in decibel-watts

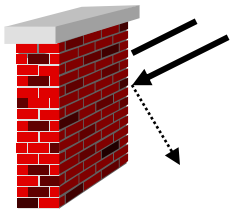
$$\begin{aligned} N &= 10 \log k + 10 \log T + 10 \log B \\ &= -228.6 \text{ dBW} + 10 \log T + 10 \log B \end{aligned}$$

Interference

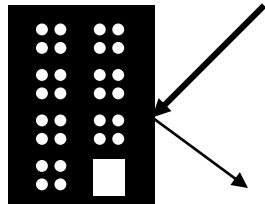
- Interference from other sources (third parties)
 - **Co-channel interference**: another sender uses the **same spectrum**
 - **Adjacent-channel interference**: another sender uses some other part of the radio spectrum, but receiver filters are not good enough to fully suppress it

Signal propagation phenomena

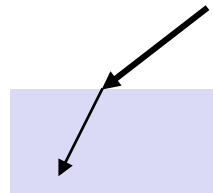
- Shadowing – obstruction of the signal
- Reflection – occurs when signal encounters a surface that is large relative to the wavelength of the signal
- Refraction – occurs when the density of a medium changes
- Scattering – scatters the incoming signal into several weaker outgoing signals. Occurs when incoming signal hits an object whose size is in the order of the wavelength of the signal or less
- Diffraction – bends the signal. Occurs at the edge of an impenetrable body that is large compared to wavelength of radio wave



shadowing



reflection



refraction



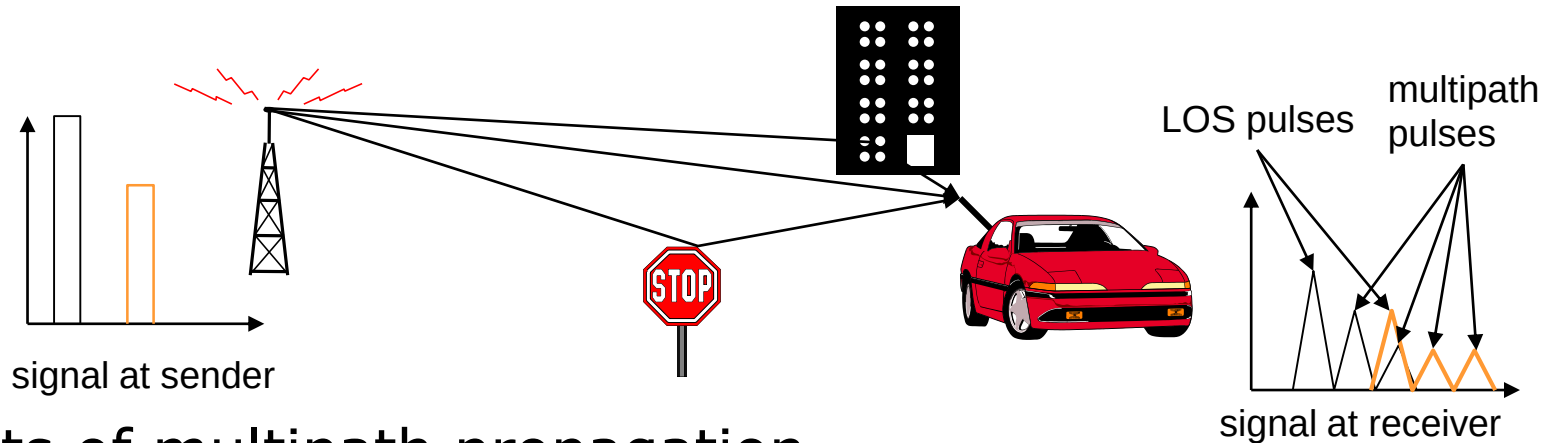
scattering



diffraction

Multipath propagation

- Signal can take many different paths between sender and receiver due to reflection, scattering, diffraction



- Effects of multipath propagation

- Fading

- Multiple copies of a signal arrive at different phases
 - If phases add destructively, the signal level declines, making detection more difficult

- Intersymbol interference (ISI)

- One or more delayed copies of a pulse may arrive at the same time as the pulse for a subsequent bit

Signal-to-noise ratios

Expression E_b/N_0

- Ratio of signal energy per bit to noise power density per Hertz

$$\frac{E_b}{N_0} = \frac{ST_b}{N_0} = \frac{S/R}{N_0} = \frac{S}{kTR}$$

S – Signal power

T_b – Bit duration

R – Bit rate

- The bit error rate for digital data is a function of E_b/N_0
- As bit rate (R) increases, transmitted signal power (S) must increase to maintain required E_b/N_0

Expression E_b/N_0

- We can relate E_b/N_0 with SNR as follows:
 - The noise in a signal with bandwidth B_T is

$$N = N_0 B_T$$

- Substituting, we have

$$\frac{E_b}{N_0} = \frac{S}{N_0 R} = \frac{S}{N} \frac{B_T}{R}$$